CMP5367:
Artificial Intelligence and Machine Learning

Lecture 4: Regression

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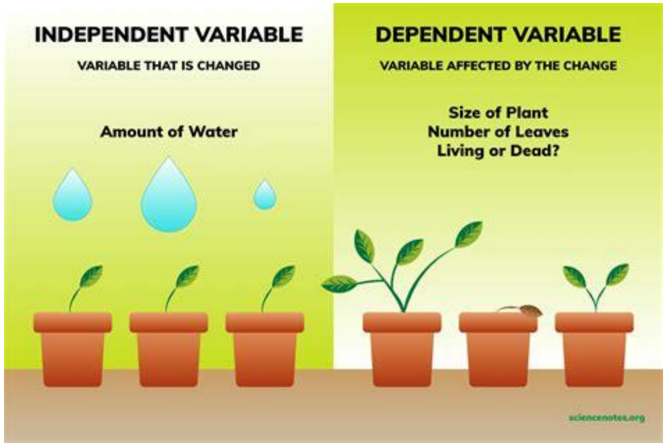
Outline

- Regression concept
- Linear regression model
- Regression analysis
  - Scatter Diagram
  - Least Square Line (random error)
  - Interpretation of a and b
  - Assumptions of the Regression Model
- Logistic Regression
- Types of regression models

What is linear regression?

Linear regression analysis is used to predict the value of a variable based on the value of another variable. The variable you want to predict is called the dependent variable. The variable you are using to predict the other variable's value is called the independent variable.



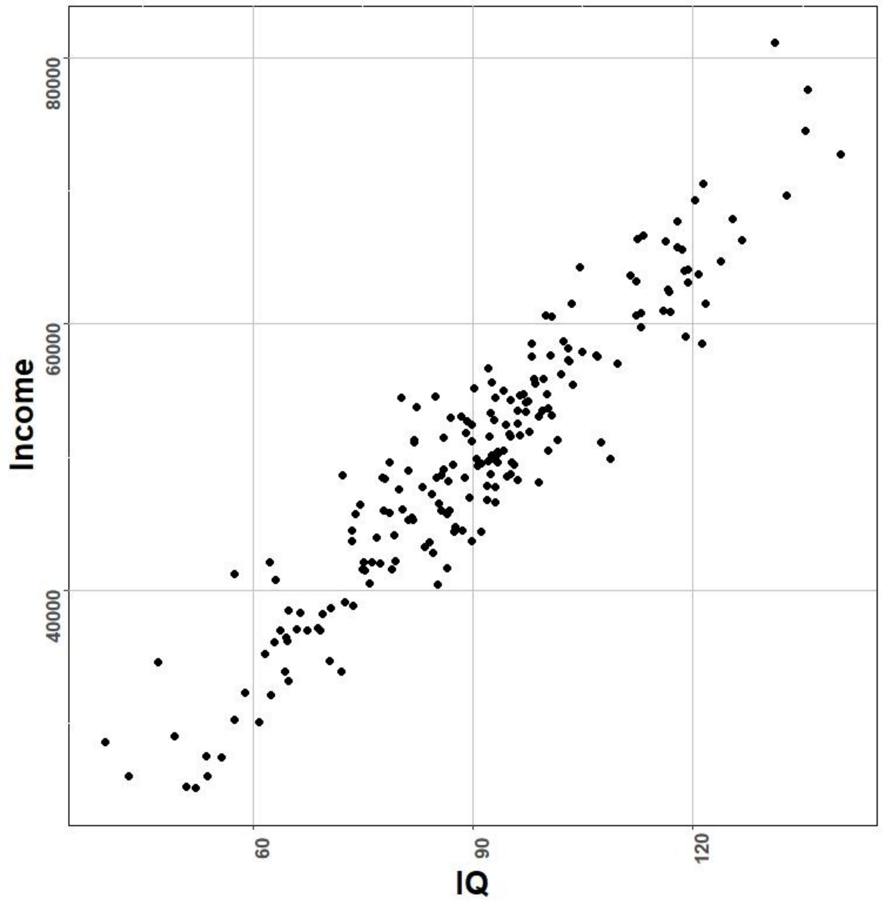
Remember:

1. LR predicts a continuous numeric value (1,2.4, 4.3, 6…etc.)
2. LR is supervised learning (I have labels for my data spam/non-spam)
3. LR is of two types: Simple (where only one independent variable is present) and Multiple (more than one independent variable).

Scatter diagram

A plot of paired observations is called a scatter diagram.

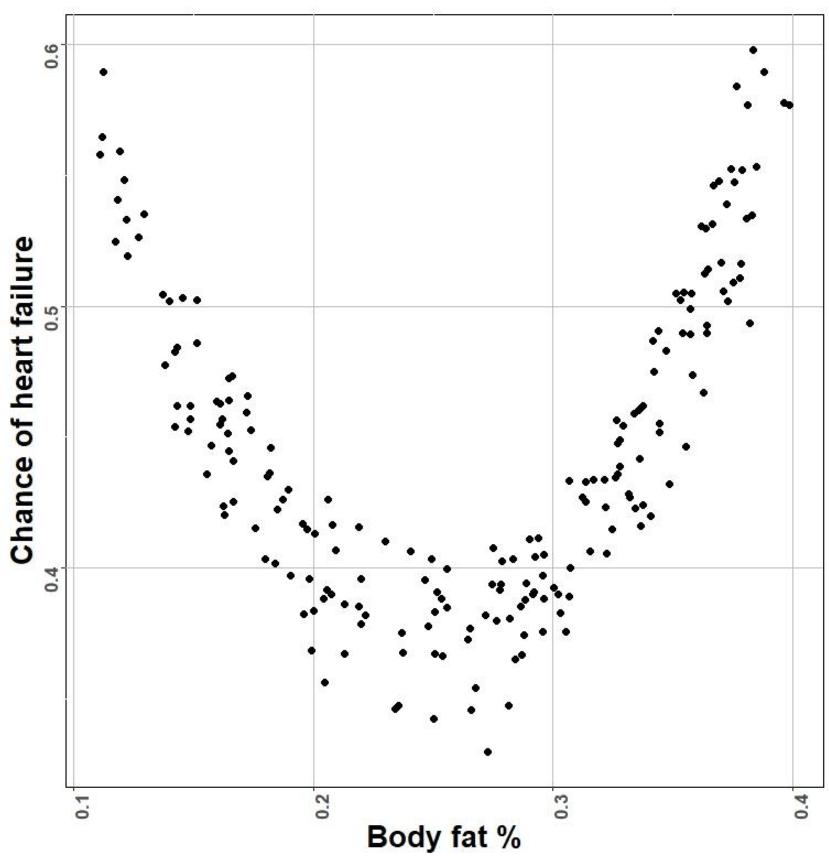
Scatter diagram will help us determine if **DV** (*dependent variable is which what I am predicting e.g. the IQ of a person)* and **IV** (*the independent variable which is his/her income*) have a linear relationship.



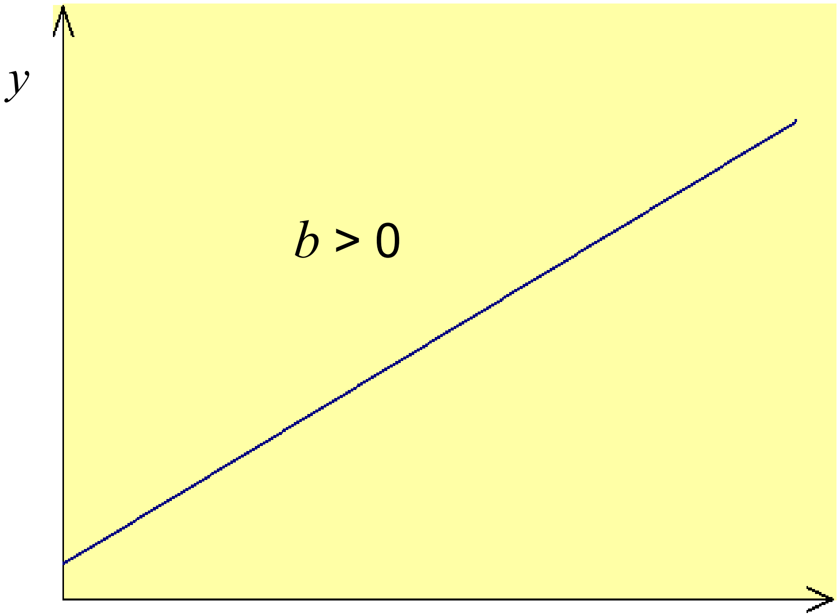
Scatter diagram

It is important to perform a scatterplot because it helps us to see if the relationship is linear.

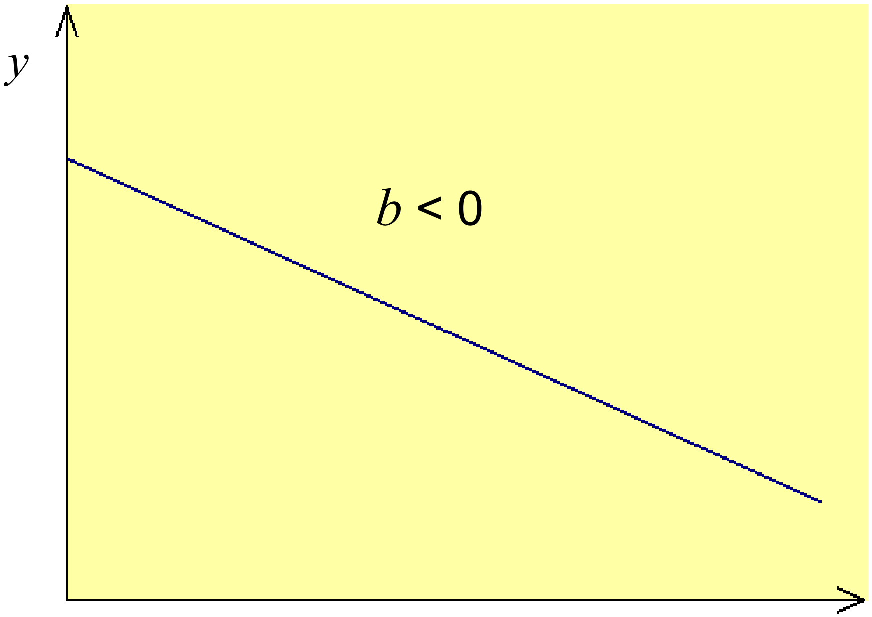
If it is not linear, it is not sensible to use linear regression instead we can opt for a non-linear Regression (Polynomial Regression)



Positive and negative linear relationships



(a) Positive linear relationship.

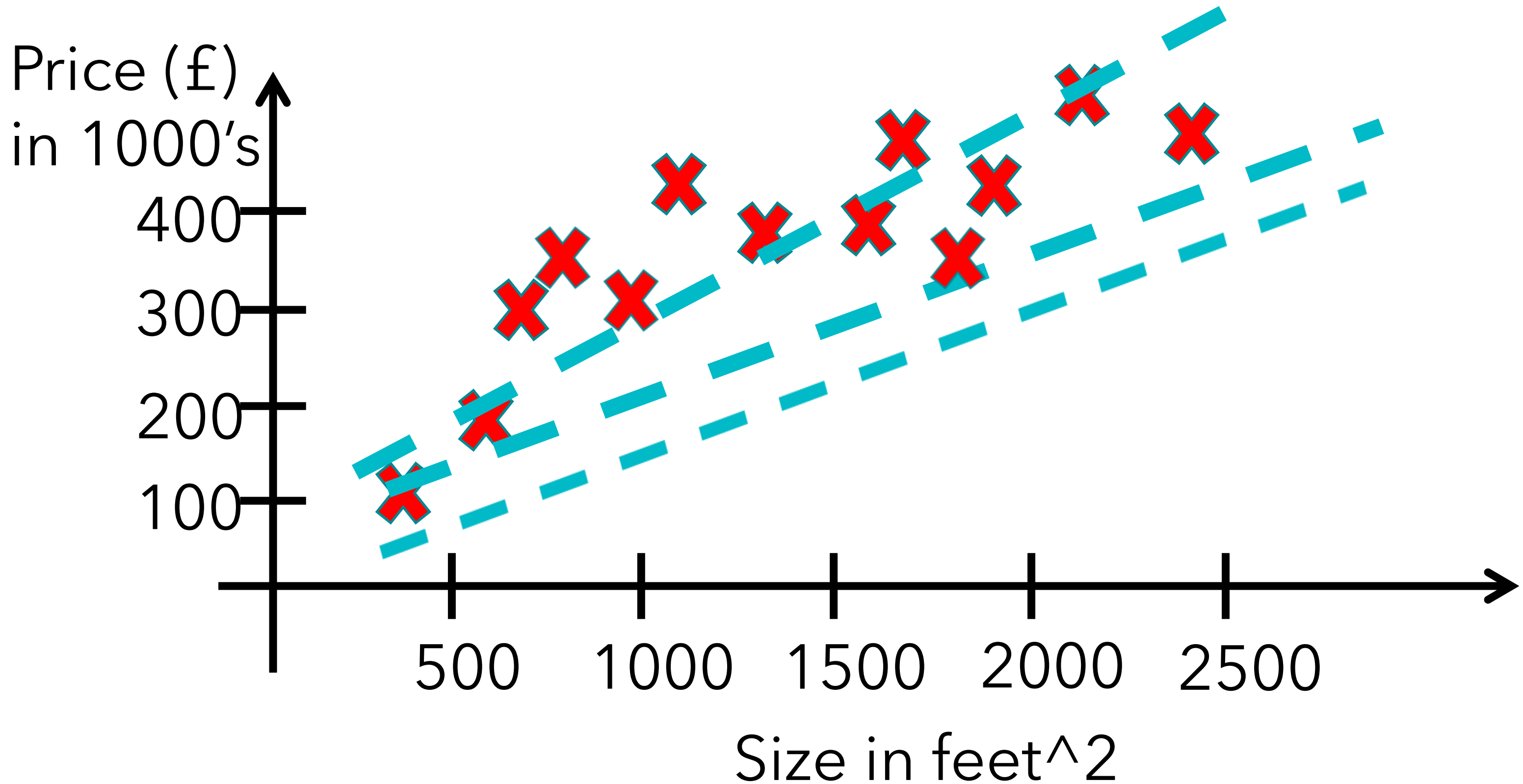


(b) Negative linear relationship.

Remember the predicting car price (DV) (it will go in negative) with respect to mileage and age (IVs)

House pricing prediction

Fit a straight line or surface that minimizes the discrepancies between predicted and actual output values.



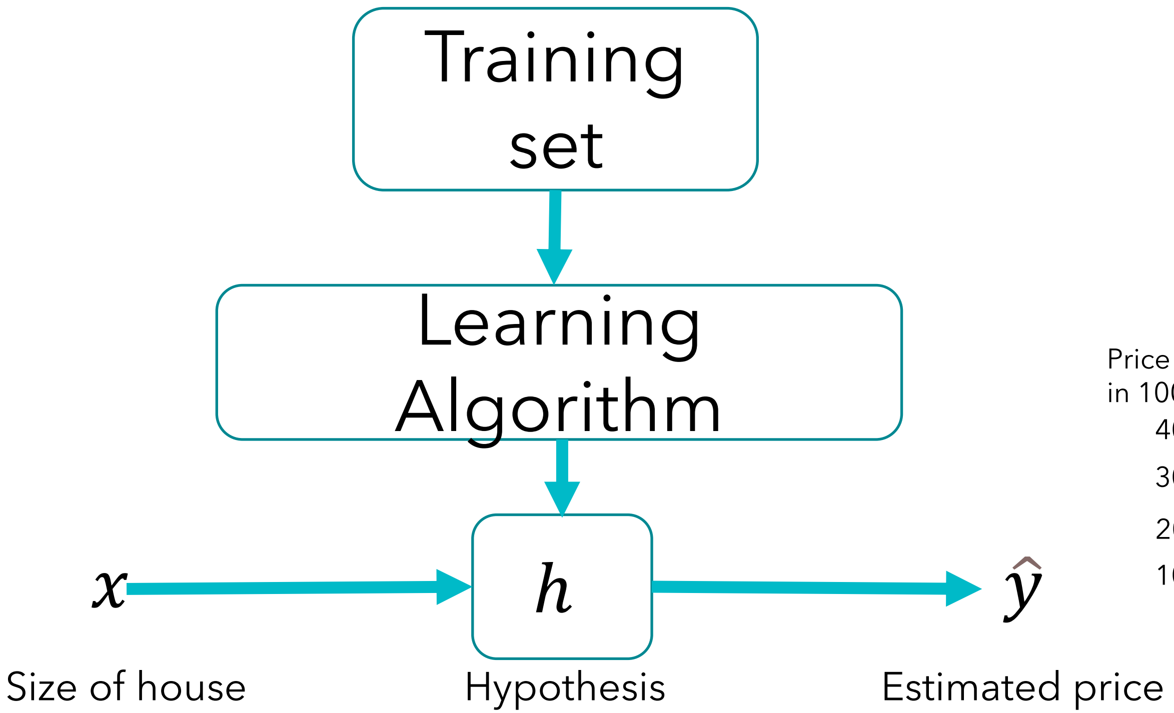
Why linear regression is important?

1. Relatively simple
2. Easy-to-interpret mathematical formula.
3. Reliably predict the future

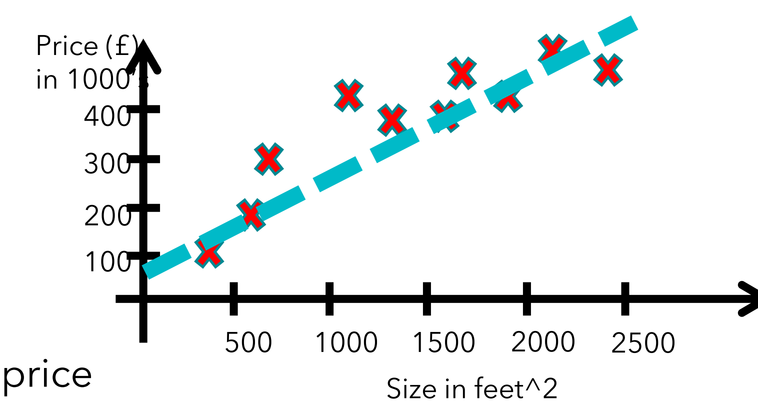
4. Linear regression can be applied to various areas in business and academic study:

    - ❑ Analyse pricing elasticity
    - ❑ Assess risk in an insurance company
    - ❑ Sports analysis
    - ❑ Likelihood of cyber threat detection

Model representation



$$\hat{y} = f(x)$$



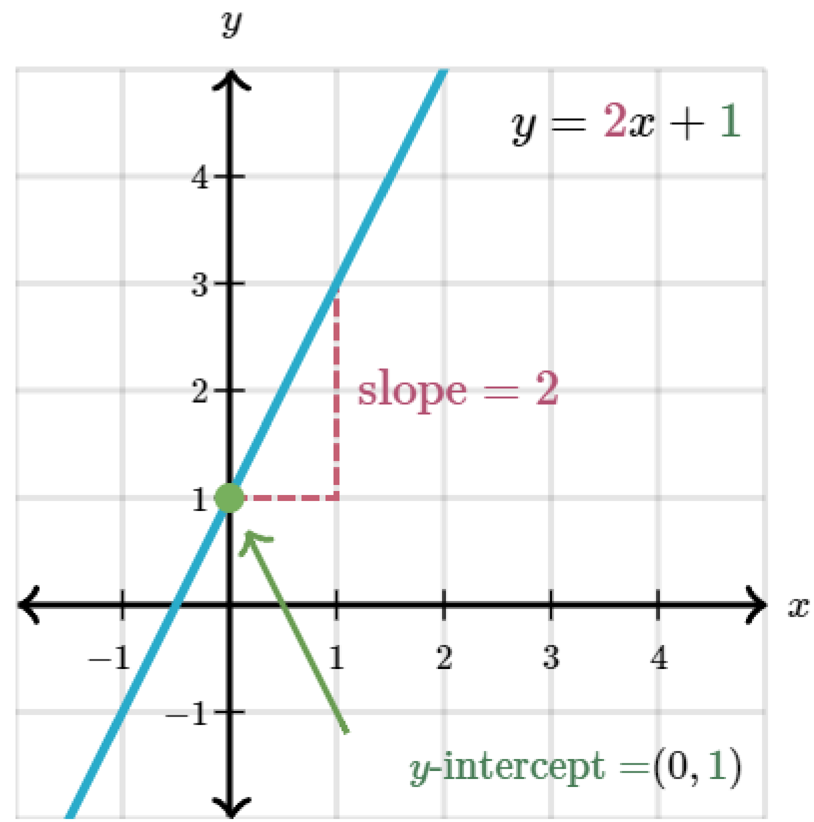
Univariate linear regression

How to define the line?

Slope-intercept equation defines two degrees of freedom for the line

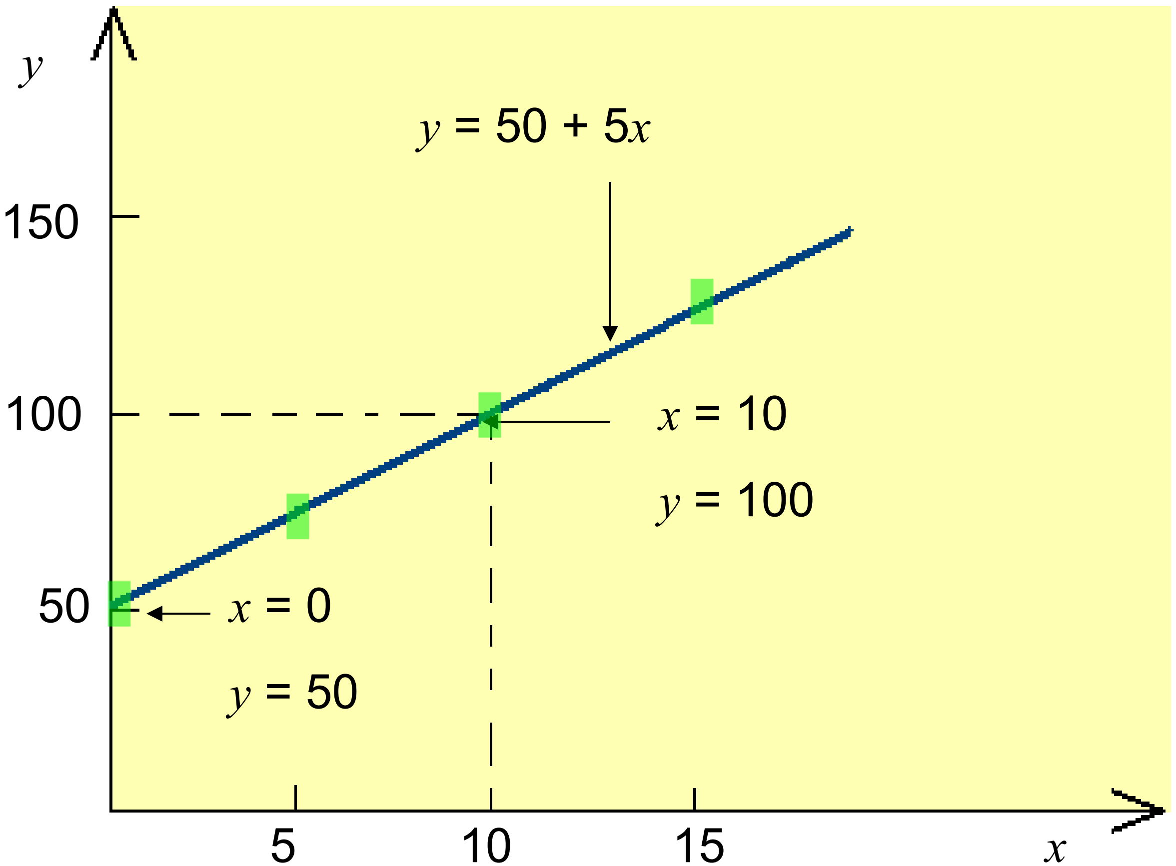
$$\hat{y} = f(x) = b + wx$$

Slope (w): is the change in ***y*** over the change of ***x*** (the angle of the line, i.e. the rate of change)
Intercept (b) : is the point where the line intercepts with the ***y*** axis, i.e. when x =0 (the position of the line on the y axis, higher or lower)
Slope and intercept are called the **coefficients** and can be referred to by other notations (e.g., m and b or a and b, etc.)



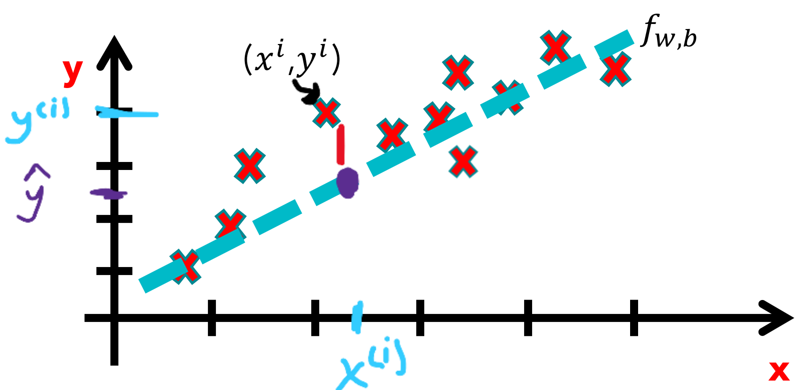
Plotting a linear equation

$$\hat{y} = w + bx$$



Main objective of linear regression

$$\hat{y} = w + bx$$



Estimates the coefficients of the linear equation that best predict the value of the dependent variable. **Minimize** the discrepancies between predicted and actual output values.

$(x^{\mathrm{i}}, y^{\mathrm{i}})$ The coordinates of **my true data** point (House 500 feet, Price 200K pounds)

$(x^i, \hat{y})$ The coordinates of **my predicted data** point. (House 500 feet, Price 180K pounds)

Objective: change the slope (w) and intercept (b) of the linear equation to minimize the distance between $\hat{\boldsymbol{y}}$ and $\boldsymbol{y}^{\boldsymbol{i}}$ for all $\boldsymbol{n}$ examples

Residual = actual y value – predicted $\hat{y}$ value

Cost Function

Mean Squared Error (MSE) cost function is used, which is the average squared error that occurred between the $\hat{y}$ **predicted** and y^{i}.

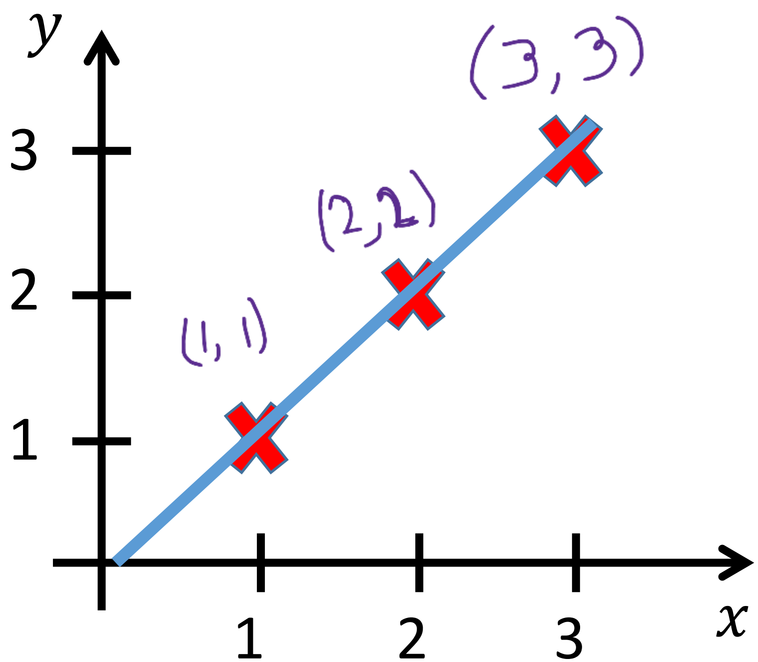
$$J(w, b) = \frac{1}{2n} \sum_{i=1}^{n} \left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

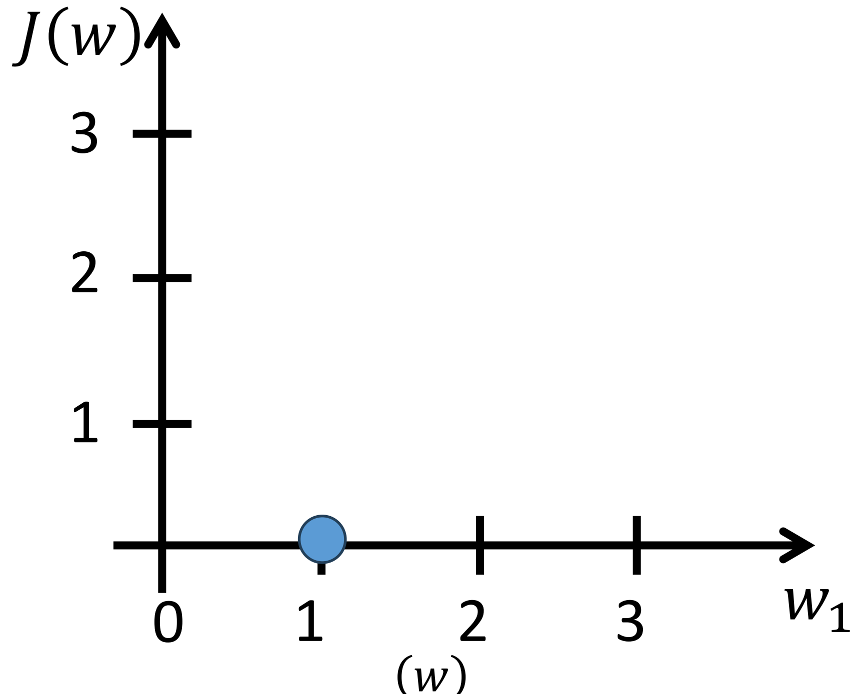
$$f(x) = b + wx = \hat{y}$$

set b to 0 for simplicity

$$f(x) = wx = \hat{y}$$

$$J(w) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$





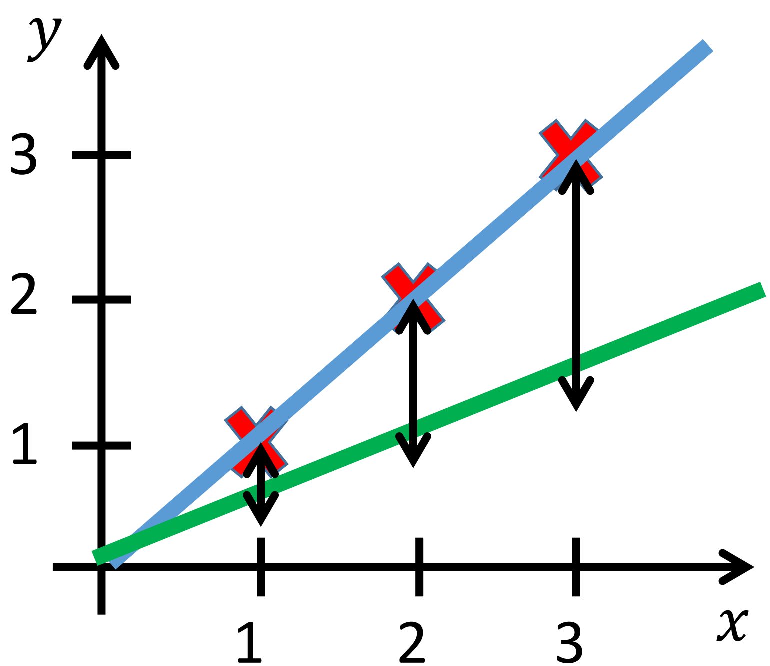
My prediction (blue line) = (1,1), (2,2), (3,3) (if w= 1)

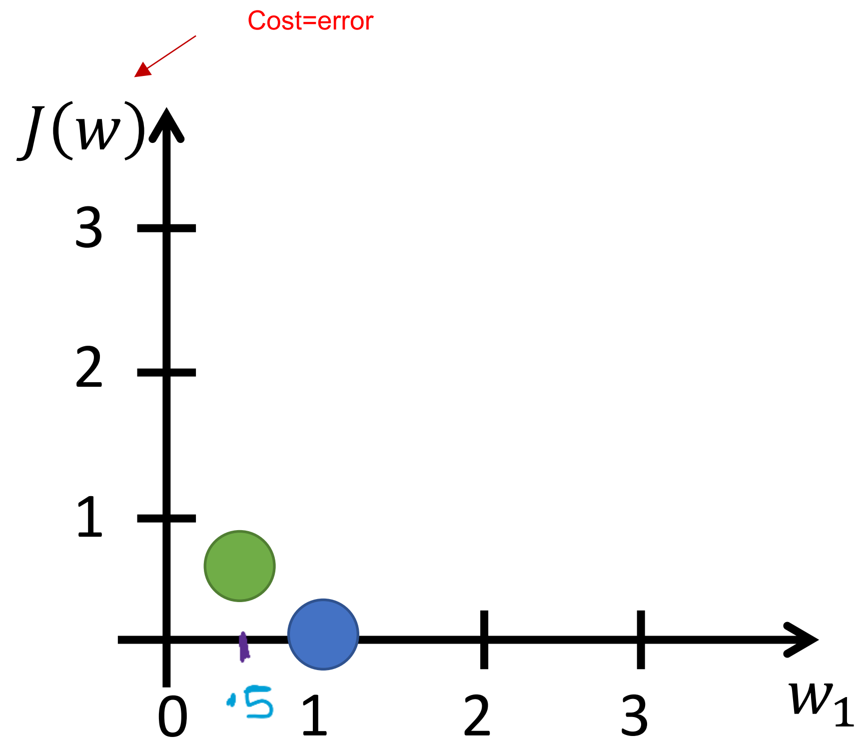
The cost is 0

No difference between my prediction and the actual

$$f(x) = wx$$

$$J(w) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$





My prediction (green line) =
(0.5,1), (1,2), (1.5,3) (if w= 0.5)

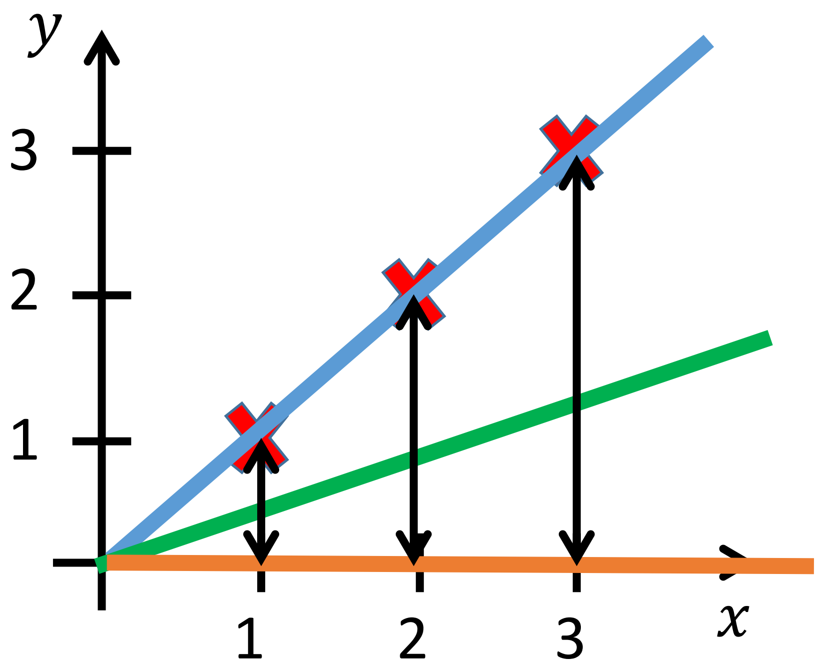
The cost =

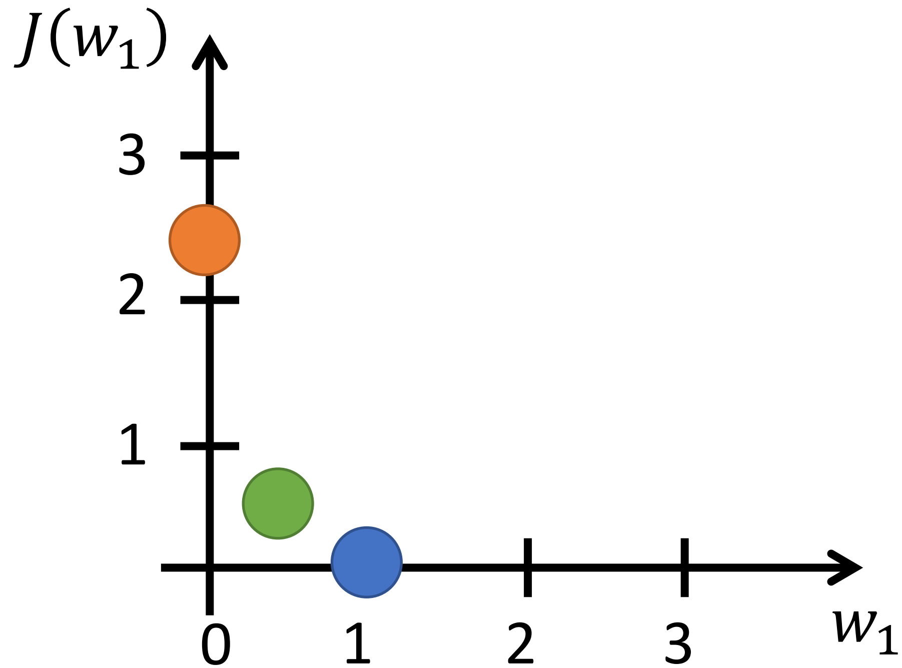
$\frac{1}{2n}\left[(0.5-1)^2 + (1-2)^2 + (1.5-3)^2\right] =$

$\frac{1}{2n} 3.5 = 3.5/6 = 0.5$

$$f(x) = wx$$

$$J(w) = \frac{1}{2n} \sum_{i=1}^{n} \left(\hat{y}^{(i)} - y^{(i)}\right)^2$$





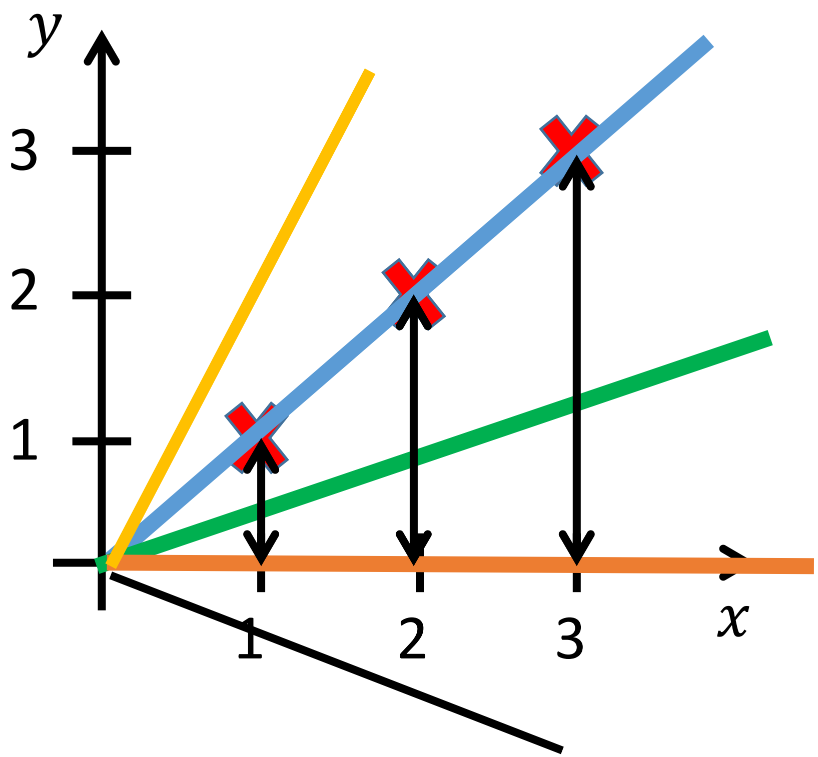
My prediction (orange line) =
(0,1), (0,2), (0,3) (if w= 0)

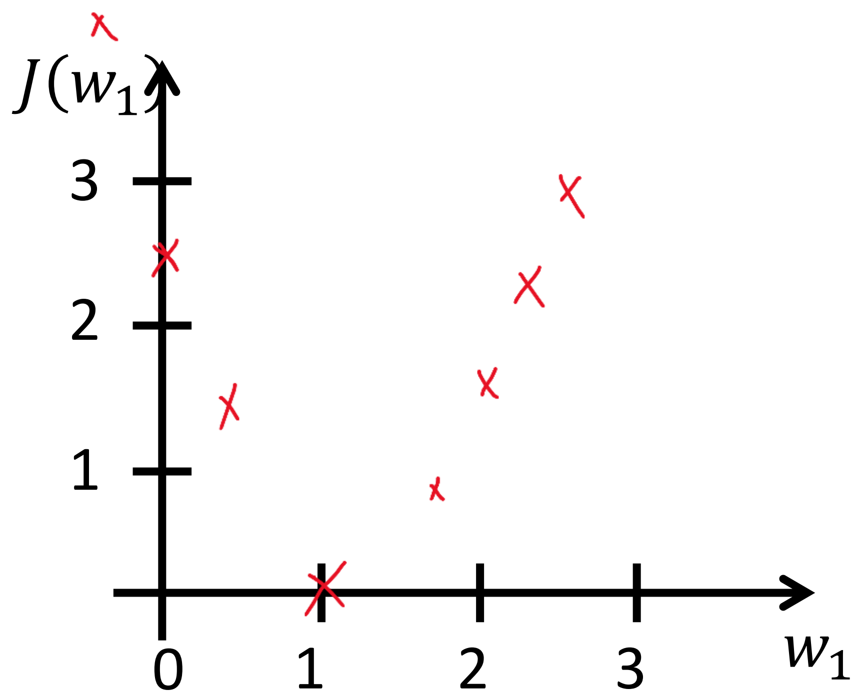
The cost:

$\frac{1}{2n}\left(1^2 + 2^2 + 3^2\right) = \frac{1}{6} 14 \approx 2.3$

$$f(x) = wx$$

$$J(w) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

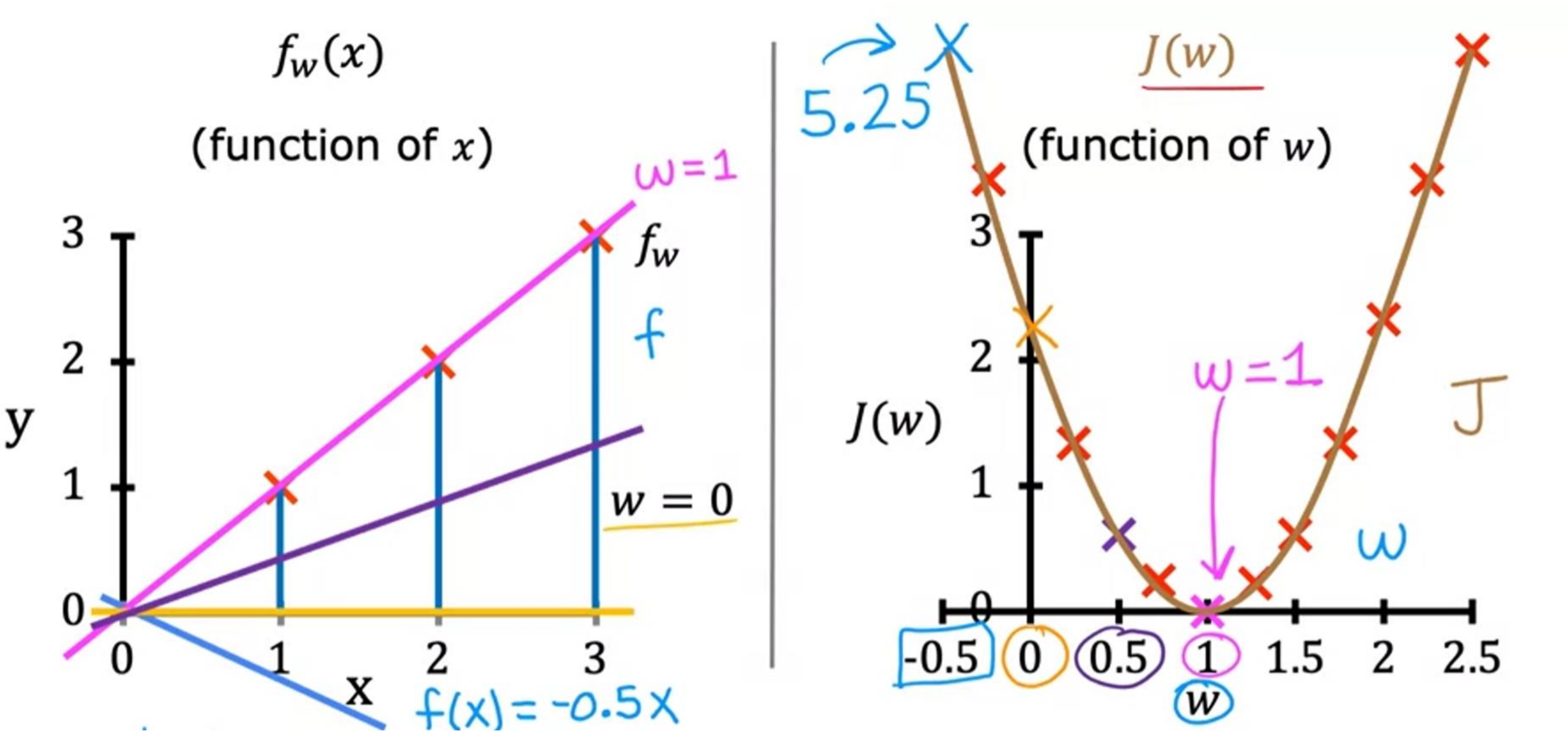




So when the cost function reaches the minima in the right graph we get the best linear model.

Objective: choose w to minimize ***J(w)***

Error graph



How to choose w and b to get the minimum error?

The least square line formula

$$w = \frac{N\,\Sigma(xy) - \Sigma x\,\Sigma y}{N\,\Sigma(x^2) - (\Sigma x)^2}$$

$$b = \frac{\Sigma y - w\,\Sigma x}{N}$$

Where:

- Σ represents a sum.
- N is the number of observations.

X and *Y* are the values in my training data

Calculate the least square line

Example: Sam found how many **hours of sunshine** vs how many **ice creams** were sold at the shop from Monday to Friday:

"x" Hours of Sunshine	"y" Ice Creams Sold
2	4
3	5
5	7
7	10
9	15

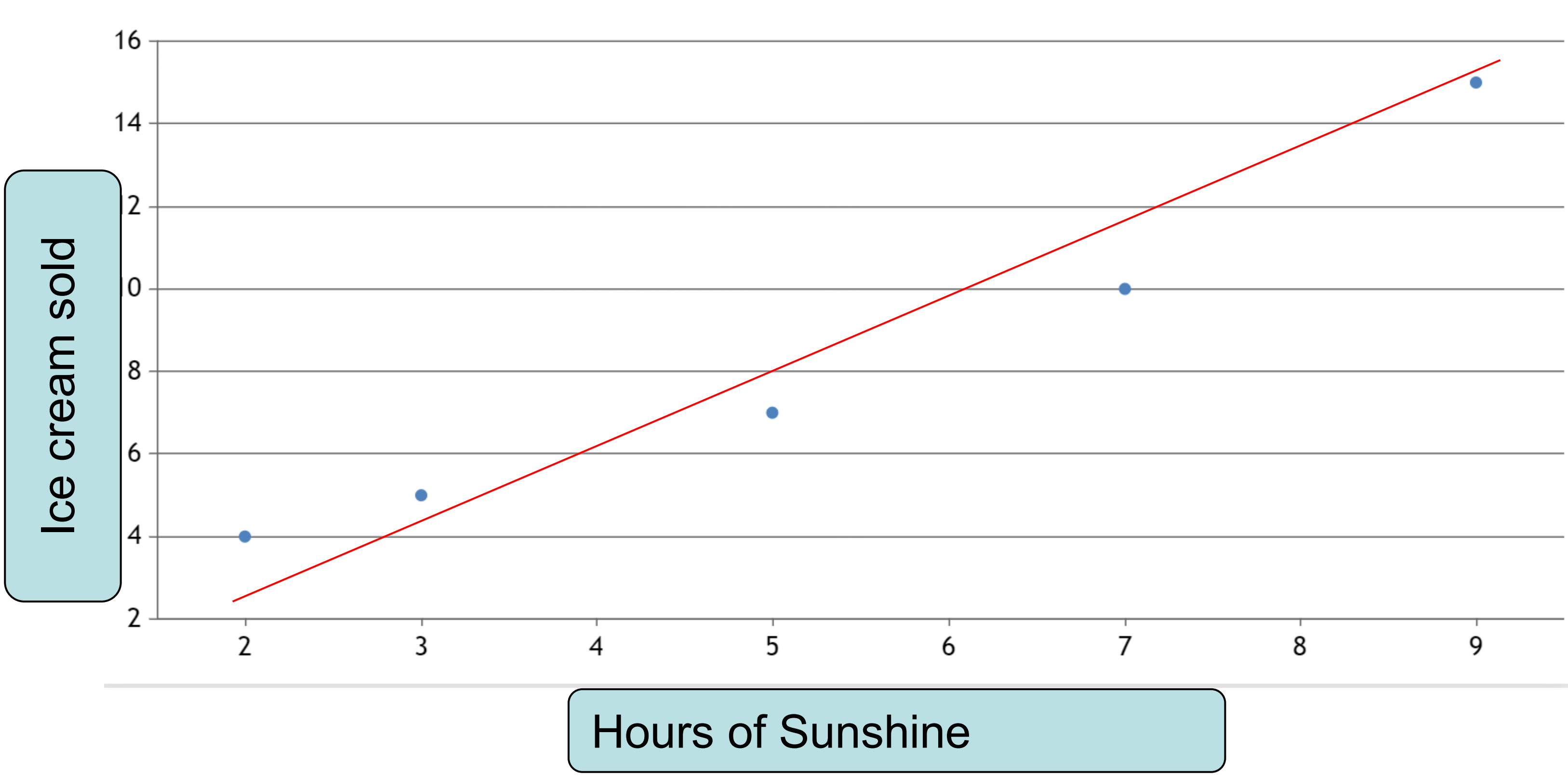
https://www.mathsisfun.com/data/least-squares-regression.html

$$w = \frac{N\,\Sigma(xy) - \Sigma x\,\Sigma y}{N\,\Sigma(x^2) - (\Sigma x)^2}$$

$$b = \frac{\Sigma y - m\,\Sigma x}{N}$$

Objective: I need to find the best slope (w) and best intercept (b) of the regression line to get the least error

$$\hat{y} = f(x) = b + wx$$



$$w = \frac{N\,\Sigma(xy) - \Sigma x\,\Sigma y}{N\,\Sigma(x^2) - (\Sigma x)^2}$$

$$b = \frac{\Sigma y - w\,\Sigma x}{N}$$

Least square error formula

Step 1: For each (x,y) calculate x^2 and xy:

x	y	x^2	xy
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135

$$W : \frac{N\,\Sigma(xy) - \Sigma x\,\Sigma y}{N\,\Sigma(x^2) - (\Sigma x)^2}$$

Step 2: Sum x, y, x^2 and xy (gives us Σx, Σy, Σx^2 and Σxy):

x	y	x^2	xy
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135
Σx: 26	**Σy: 41**	**Σx^2: 168**	**Σxy: 263**

Also **N** (number of data values) = **5**

Step 2: Sum x, y, x^2 and xy (gives us Σx, Σy, Σx^2 and Σxy):

x	y	x^2	xy
2	4	4	8
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Σx: 26	**Σy: 41**	**Σx^2: 168**	**Σxy: 263**

Also **N** (number of data values) = **5**

Step 3: Calculate Slope *w*

$$w = \frac{N\,\Sigma(xy) - \Sigma x\,\Sigma y}{N\,\Sigma(x^2) - (\Sigma x)^2}$$

Step 3: Calculate Slope w

$$w = \frac{N\ \Sigma(xy) - \Sigma x\ \Sigma y}{N\ \Sigma(x^2) - (\Sigma x)^2}$$

$$= \frac{5 \times 263 - 26 \times 41}{5 \times 168 - 26^2}$$

$$= \frac{1315 - 1066}{840 - 676}$$

$$= \frac{249}{164} = \mathbf{1.5183...}$$

Step 2: Sum x, y, x^2 and xy (gives us Σx, Σy, Σx^2 and Σxy):

x	y	x^2	xy
2	4	4	8
3	5	9	15
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7	10	49	70
9	15	81	135
Σx: 26	**Σy: 41**	**Σx^2: 168**	**Σxy: 263**

Also **N** (number of data values) = **5**

Calculate Intercept **b**:

$$\mathbf{b} = \frac{\Sigma y - w\,\Sigma x}{N}$$

$$= \frac{41 - 1.5183 \text{ x } 26}{5}$$

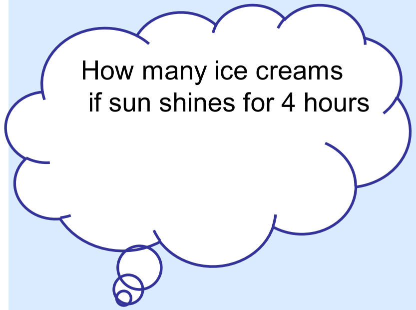
$$= \mathbf{0.3049...}$$

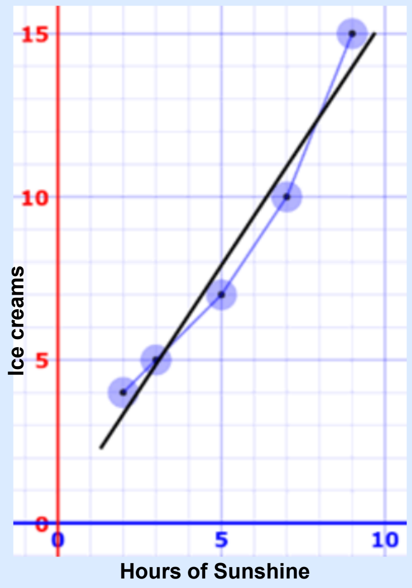
$y = 1.518x + 0.305$

Let's see how well it works:

x	y	y = 1.518x + 0.305	error
2	4	3.34	−0.66
3	5	4.86	−0.14
5	7	7.89	0.89
7	10	10.93	0.93
9	15	13.97	−1.03

Here are the (x,y) points and the line **y = 1.518x + 0.305** on a graph:





Nice fit!

Any Questions so far?

If No

Then

Quiz

Linear Regression:
Multiple independent random variables

- Multiple Linear Regression (MLR): Two or more (attribute values) as Inputs (Independent Variables) and Output (Dependent Variable) are all numeric.
- Output is the sum of the weighted input attribute values.
- The trick is to find good values for the weights (constants).

Simple Linear Regression

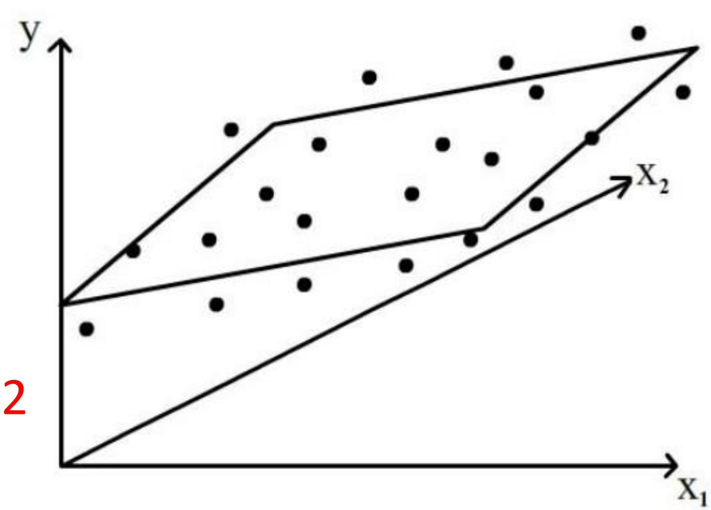
$$y = b_0 + b_1 * x_1$$

Multiple Linear Regression

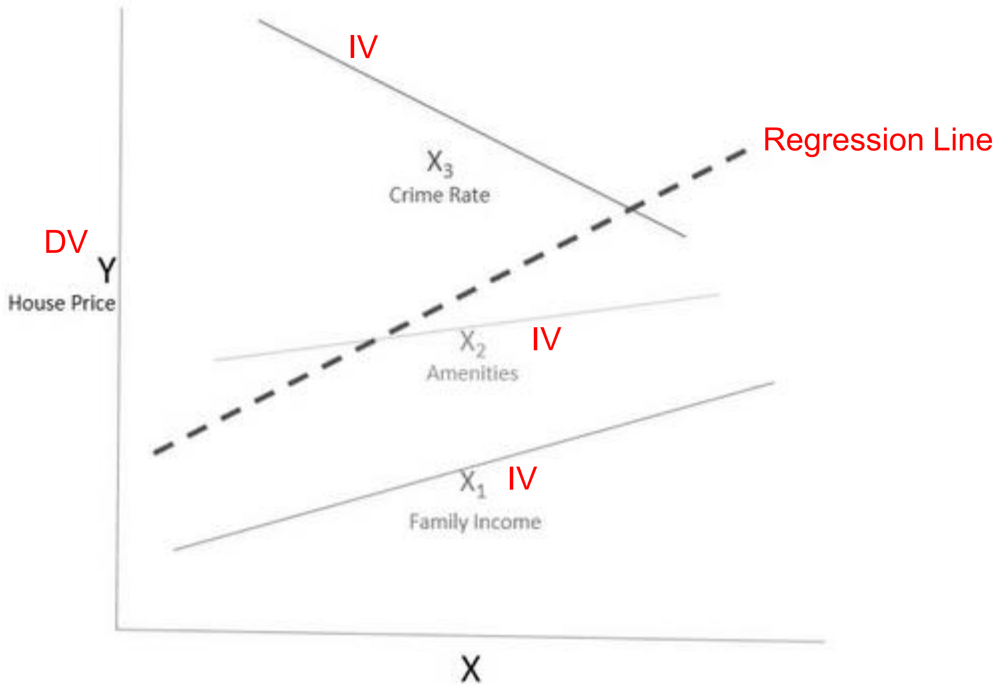
Dependent variable (DV) Independent variables (IVs)

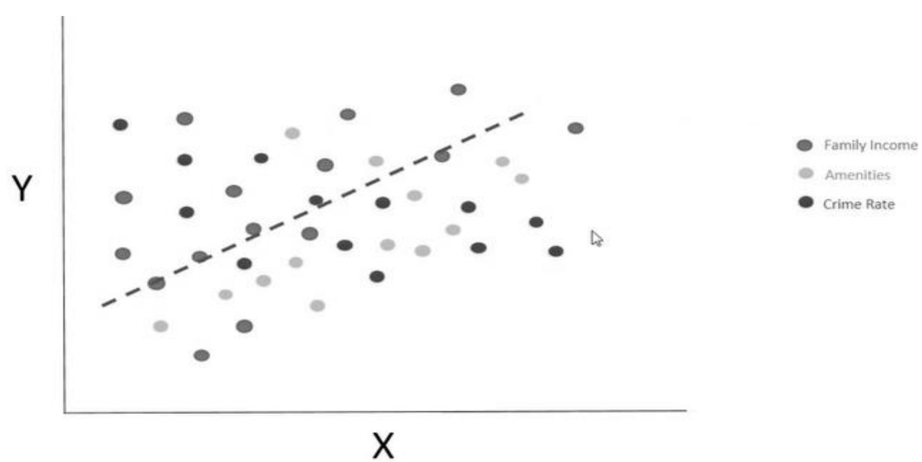
$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \ldots + b_n * x_n$$

y = b (intercept) + slope x1+ slope x2



Linear Regression:
Multiple independent random variables





Linear Regression: Multiple independent random variables

- Work most naturally with numeric attributes

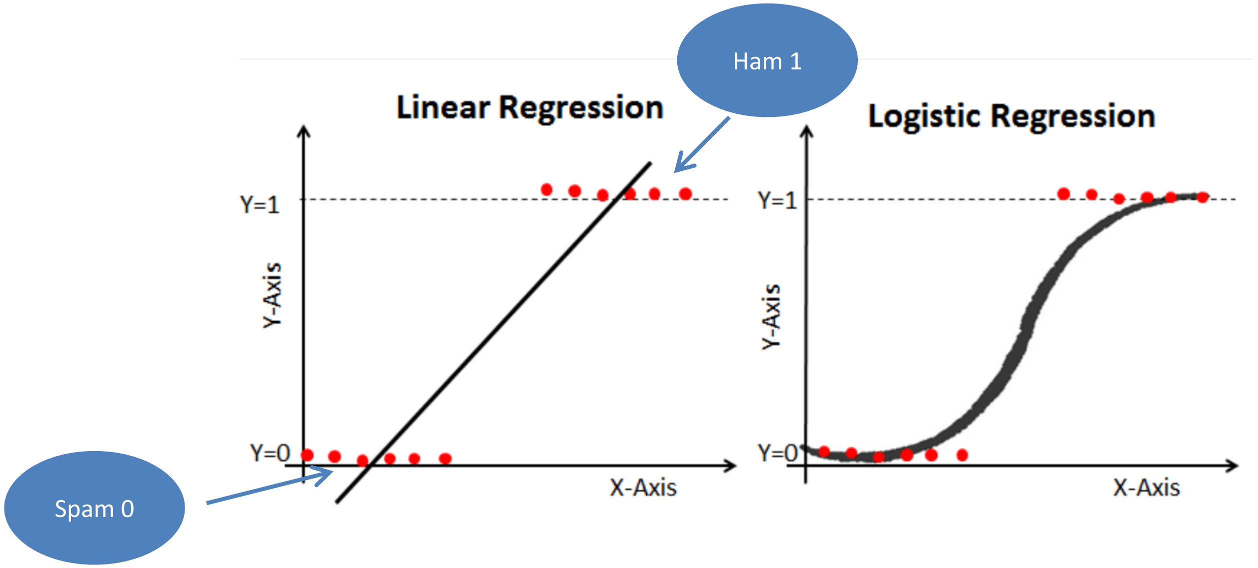
- Standard technique for numeric prediction
  - Outcome is linear combination of attributes

$$x = w_0 + w_1 a_1 + w_2 a_2 + \cdots + w_k a_k$$

- Weights are calculated from the training data

Logistic Regression (classification problem):

binary classification problem in which y can take on only two values, 0 and 1 classification (spam and ham).



To avoid this problem, we must model probability of X (*p(X))*(i.e. our prediction) using a function that gives outputs between 0 and 1 for all values of *X (the attributes/Features)*. Many functions meet this description. In logistic regression, we use the logistic function.

Logistic Regression predicts probabilities not raw values

Logistic Function

Remember our linear regression was with the formula:

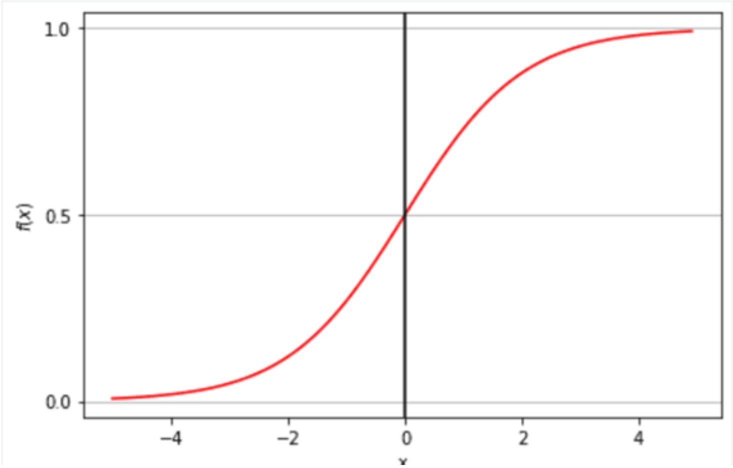
$$\hat{y} = w + bx$$

We change *w* and *b* to get the best line fit.

In logistic regression we change this to the logistic function to have the model prediction value between 0 and 1 to fit our binary classification (0 good mail, 1 spam)

Logistic Function is:

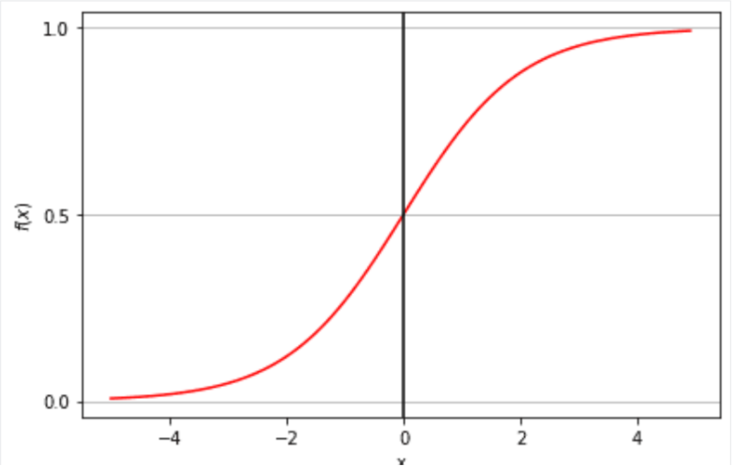
$$\hat{y} = \frac{e^{w+bx}}{1+e^{w+bx}}$$



e, “Euler’s number”, 2.718.

Sigmoid function (logistic function):

- The sigmoid function gives an ‘S’ shaped curve that can take any real-valued number and map it into a value between 0 and 1.
- If the output of the sigmoid function is more than 0.5, we can classify the outcome as 1 or YES, and if it is less than 0.5, we can classify it as 0 or NO. For example, *if the output is 0.75, we can say in terms of the probability that there is a 75 percent chance that a patient will suffer from cancer.*
- *We set the threshold based on the problem*
- *What do you think a good threshold for detecting the probability of a customer will not pay back the loan?*



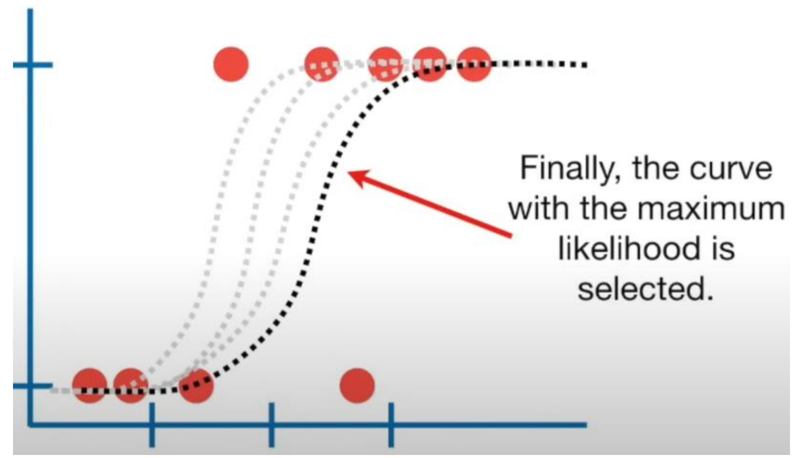
Maximum Likelihood function (MLE)

$$\hat{y} = \frac{e^{w+bx}}{1+e^{w+bx}}$$

$$f(\omega, b) = \prod_{i:y_i=1} (px_i) \prod_{i:y_i=0} \left(1 - p(x_i)\right)$$

Objective in training
is to maximise the likelihood function **(Maximum Likelihood Estimation) MLE.**

The MLE is a "likelihood" maximization method, while least squares is a distance-minimizing approximation method. Maximizing the likelihood function determines the parameters that are most likely to produce the observed data.



“*Maximum likelihood estimation in logistic regression finds the weights that make the model’s predicted probabilities most consistent with the actual labels.*

It’s like tuning the model to say:

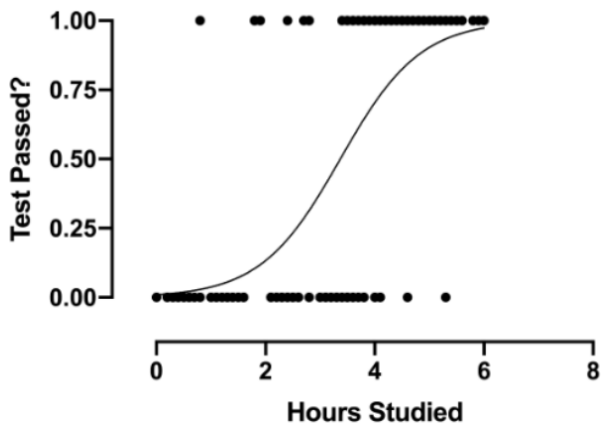
‘Given what I’ve seen, these weights make my predictions most believable.’”

What type of ML predictions can be trained by Logistic Regression?

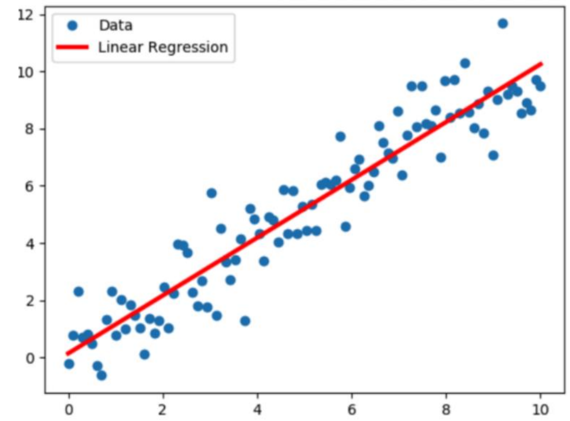
Types of Logistic Regression

- **Binary Logistic Regression**:
  Spam or Not Spam, Cancer or No Cancer.
- **Multinomial Logistic Regression**:
  three or more nominal categories, such the class a student will choose.
- **Ordinal Logistic Regression**:
  three or more ordinal categories such as rating   from 1 to 5.

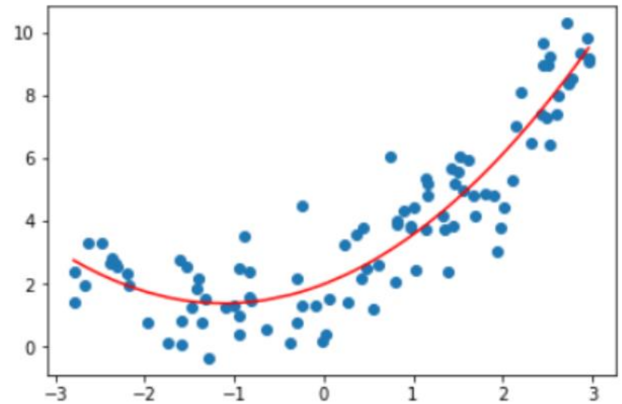
Different types of regression models



Logistic Regression



Linear Regression



Polynomial Regression

and others …

POPULAR REGRESSION ALGORITHMS

1. Linear Regression
2. Ridge Regression ((independent variables are highly correlated).
3. Lasso Regression
4. Support Vector Regression (SVR)
5. Decision Tree Regression 5.
6. Random Forest Regression 6.
7. Logistic Regression (for binary classification)

Note the difference between regression algorithms and regression problems:

➢ logistic regression is used for binary classification problems, not regression

➢ Some algorithms (e.g. decision trees, SVM, neural networks) can be used to solve regression problems

Summary (What we learned today)

- What is Linear Regression, in one line?
- Dependent and Independent Variables
- Equation for Linear Regression
- How to get w
- How to get b
- Pseudocode for Linear Regression
- Multiple Linear Regression